

Claims

1. A pharmaceutical composition for use in preventing and treating diabetic complications comprising—consisting essentially of the extract of
5 *Homonoia riparia* Lour. or the fraction thereof,

wherein the diabetic complication is selected from the group consisting of diabetic retinopathy, diabetic cataract, diabetic nephropathy, diabetic
10 neuropathy, heart disease, cancer, osteoporosis, and atherosclerosis,

wherein the extract of *Homonoia riparia* Lour. is characteristically extracted by using mixture of water and ethanol, and

15 wherein the mixture contains water at the concentration of 0.1% ~ 50%.

2. A health functional food for use in preventing and improving diabetic complications comprising
20 consisting essentially of the extract of *Homonoia riparia* Lour. or the fraction thereof,

wherein the diabetic complication is selected from the group consisting of diabetic retinopathy, diabetic cataract, diabetic nephropathy, diabetic
25 neuropathy, heart disease, cancer, osteoporosis, and atherosclerosis,

wherein the extract of *Homonoia riparia* Lour. is characteristically extracted by using mixture of water and ethanol, and

wherein the mixture contains water at the
5 concentration of 0.1% ~ 50%.

3. A functional feed additive for use in preventing and improving diabetic complications ~~comprising~~ consisting essentially of the extract of
10 *Homonoia riparia* Lour. or the fraction thereof,

wherein the diabetic complication is selected from the group consisting of diabetic retinopathy, diabetic cataract, diabetic nephropathy, diabetic neuropathy, heart disease, cancer, osteoporosis, and
15 atherosclerosis,

wherein the extract of *Homonoia riparia* Lour. is characteristically extracted by using mixture of water and ethanol, and

wherein the mixture contains water at the
20 concentration of 0.1% ~ 50%.

~~4. A pharmaceutical composition for use as in claim 1, wherein the extract of *Homonoia riparia* Lour. is characteristically extracted by using water, C₁ ~ C₄~~
25 ~~lower alcohol, or the mixture thereof.~~

~~5. A pharmaceutical composition for use as in claim 4, wherein the lower alcohol is ethanol or methanol.~~

5 ~~6. A pharmaceutical composition for use as in claim 4, wherein the lower alcohol contains water at the concentration of 0.1% ~ 50%.~~

74. A pharmaceutical composition for use as in
10 claim 1, wherein the extract of *Homonoia riparia* Lour.
is prepared by using leaves or twigs of *Homonoia riparia* Lour.

85. A pharmaceutical composition for use as in
15 claim 1, wherein the fraction is one of those selected
from the group consisting of normal-hexane fraction,
ethyl acetate fraction, normal-butanol fraction, and
water fraction which were obtained stepwise by a
serial line fractionation performed with the extract
20 using normal-hexane, ethyl acetate, normal-butanol,
and water in that order after suspending the extract
of *Homonoia riparia* Lour. in water.

~~9. A pharmaceutical composition for use as in
25 claim 1, wherein the diabetic complications is
selected from the group consisting of diabetic
retinopathy, diabetic cataract, diabetic nephropathy,~~

~~diabetic neuropathy, heart disease, cancer, osteoporosis, and atherosclerosis.~~

~~10. A health functional food for use as in claim
5 2, wherein the extract of *Homonoia riparia* Lour. is
characteristically extracted by using water, C₁ ~ C₄
lower alcohol, or the mixture thereof.~~

~~11. A health functional food for use as in claim
10 10, wherein the lower alcohol is ethanol or methanol.~~

~~12. A health functional food for use as in claim
10, wherein the lower alcohol contains water at the
concentration of 0.1% ~ 50%.~~

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~~136. A health functional food for use as in claim
2, wherein the extract of *Homonoia riparia* Lour. is
prepared by using leaves or twigs of *Homonoia riparia*
Lour.~~

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~~147. A health functional food for use as in claim
2, wherein the fraction is one of those selected from
the group consisting of normal-hexane fraction, ethyl
acetate fraction, normal-butanol fraction, and water
25 fraction which were obtained stepwise by a serial line
fractionation performed with the extract using normal-
hexane, ethyl acetate, normal-butanol, and water in~~

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that order after suspending the extract of *Homonoia riparia* Lour. in water.